

**Development of a Quantitative Susceptibility Mapping and
Susceptibility Source Separation Pipeline for Ex Vivo Spinal
Cord Imaging**

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Dear Pat Sullivan:

During this work term (Engineering 004W), I was employed as a Research Intern at Polytechnique Montréal. This was my first work term at Polytechnique, and I was supervised by Dr. Eva Alonso Ortiz, Director at the NeuroPoly Laboratory.

The NeuroPoly Laboratory specializes in magnetic resonance imaging processing, specifically the development of quantitative imaging methods that identify biomarkers associated with neurodegenerative diseases such as Multiple Sclerosis. During this work term, I developed and validated a software pipeline to produce quantitative susceptibility maps from 2D spinal cord data.

The enclosed work report titled *Development of a Quantitative Susceptibility Mapping and Susceptibility Source Separation Pipeline for Ex Vivo Spinal Cord Imaging* resulted from a request from the Co-op office. The purpose of this report is to document my research during my internship and to serve as a point of reference for the future development of quantitative susceptibility mapping of the spinal cord, as this is one of the lab's key areas of interest.

If there are any questions concerning this report, I would be pleased to discuss them with you.

Yours truly,

Sam Ruttgaizer

Development of a Quantitative Susceptibility Mapping and Susceptibility Source Separation Pipeline for Ex Vivo Spinal Cord Imaging

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Executive Summary

Quantitative susceptibility mapping is a magnetic resonance imaging postprocessing technique that outputs a map of an imaged tissue's susceptibility and, in turn, gives an estimate of the diamagnetic myelin and paramagnetic iron content of the regions mapped. In the brain and spinal cord, neurodegenerative diseases such as multiple sclerosis cause myelin loss and iron accumulation. Quantitative susceptibility mapping gives direct insight into the progression and identification of neurodegenerative diseases that cause these effects. While many methods for this technique were initially developed for use in the brain, the spinal cord is only now being explored because of its small size and position in the body.

The purpose of this report is to provide evidence that quantitative susceptibility mapping is viable in the spinal cord by developing and analyzing a software pipeline that performs susceptibility source separation, producing positive and negative susceptibility maps for an ex vivo spinal cord sample with matching histology data for accuracy assessment, as well as a myelin water fraction measurement from a second acquisition, which provided an additional point of comparison.

The histology and myelin water fraction correlated strongly. Of the two susceptibility source separation methods tested, the anisotropic method agreed more closely with both myelin references than the isotropic method, which was unable to separate the positive and negative susceptibility components in this dataset.

Future work should include diffusion imaging, testing on additional specimens, and in vivo studies to compare against ex vivo references.

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List of Acronyms

ANISO-QSM - Anisotropic Nonlinear Inversion for Susceptibility-separation Optimization

ANTs - Advanced Normalization Tools

B0 - Main static magnetic field

CNR - Contrast-to-Noise Ratio

DWI - Diffusion-Weighted Imaging

FA - Fractional Anisotropy

FOV - Field of View

GM - Grey Matter

GUI - Graphical User Interface

HPF - High-Pass Filter

MEDI - Morphology Enabled Dipole Inversion

meGRE - Multi-Echo Gradient-Recalled Echo

MESE - Multi-Echo Spin-Echo

MRI - Magnetic Resonance Imaging

MS - Multiple Sclerosis

MVF - Myelin Volume Fraction

MWF - Myelin Water Fraction

PBS - Phosphate-Buffered Saline

PFA - Paraformaldehyde

QSM - Quantitative Susceptibility Mapping

R2 - Transverse relaxation rate ($1/T_2$)

R2* - Effective transverse relaxation rate ($1/T_2^*$)

R2' - Reversible transverse relaxation rate ($R_2^* - R_2$)

ROI - Region of Interest

SCT - Spinal Cord Toolbox

SEPIA - Susceptibility mapping Pipeline tool for phase images

SMV - Spherical Mean Value

STI - Susceptibility Tensor Imaging

T2 - Transverse relaxation time

T2* - Effective transverse relaxation time

TE - Echo Time

TKD - Thresholded K-space Division

TR - Repetition Time

V-SHARP - Variable-kernel Sophisticated Harmonic Artifact Reduction for Phase data

WM - White Matter

1 Introduction and Background

At NeuroPoly, one of the lab's main research topics is quantitative susceptibility mapping (QSM) of the brain and spinal cord in magnetic resonance imaging (MRI). QSM image processing is a technique that quantifies a tissue's magnetic susceptibility from MRI images. This is particularly useful, as it enables the creation of images that show meaningful data reflecting the tissue's chemical composition. Because different tissues have characteristic susceptibility values, these maps can be used to identify specific compounds such as iron and myelin [1]. Traditional MRI images show contrast between different tissue types and allow a radiologist to make a clinical diagnosis based on the geometry and structures they see, but this diagnosis is qualitative because the chemical makeup of the tissues cannot be precisely determined. QSM has been validated in the brain as a technique for identifying iron and myelin and is particularly useful for detecting and tracking multiple sclerosis (MS), as iron accumulation and demyelination are key signs of the disease's progression [2].

Building on QSM, new methods have been developed to separate the positive and negative elements of a susceptibility map, a process known as susceptibility source separation. Because QSM maps provide a bulk susceptibility value for each voxel, positive and negative contributions from different tissue types may cancel out, leaving the exact tissue makeup undetermined. Susceptibility source separation creates two maps, one showing the positive susceptibility of the tissue, closely related to the amount of iron, and the other the negative, related to myelin. This allows even finer structural detail to be produced and identified in these maps [3].

While QSM and susceptibility source separation have been extensively applied in the brain, and software pipelines exist to generate these maps, the spinal cord is a new area of focus for these techniques. This report will cover the development of a QSM data-processing pipeline and apply susceptibility source separation methods to an ex vivo (tissue studied outside a living organism) spinal cord dataset that also includes histology (microscopic tissue analysis) images. These images will be used to validate the accuracy of the processed data.

The report will review existing methods and algorithms for brain QSM and source separation, and present results from applying these techniques to the spinal cord. Statistical analysis will be performed to compare the susceptibility source separation maps with other quantitative techniques, such as myelin water fraction (MWF), and with histology data to assess accuracy.

This report will provide a record of the work done on this dataset for future interns and researchers at NeuroPoly to build on. It will also contribute to the field of MRI research by providing insights into the accuracy of susceptibility source separation applied to the spinal cord, and by detailing a usable pipeline that others in the field can adapt for other spinal cord data.

Spinal cord imaging, done primarily with MRI, is critical for the diagnosis of the type and extent of a spinal cord injury, whether that be trauma-caused or non-traumatic, such as neurodegenerative diseases. The spinal cord is a small structure with a unique geometry that makes in vivo analysis difficult, especially for emerging quantitative analysis methods that build on MRI-based qualitative diagnoses to provide more information on specific

spinal cord makeup, allowing for improved decision-making and clarity in treatment decisions [4].

1.1 Spinal Cord Anatomy

The spinal cord and brain clearly differ in external geometry: the brain is large and spherical, whereas the cord is long and narrow, similar to a rope. However, more critical differences are important to highlight in the internal makeup and in the external features surrounding each organ.

The spinal cord's characteristic internal shape is best described as a butterfly pattern that distinguishes the grey matter (GM), surrounded by white matter (WM), which forms an external circular shape, as can be seen in Figure 1.

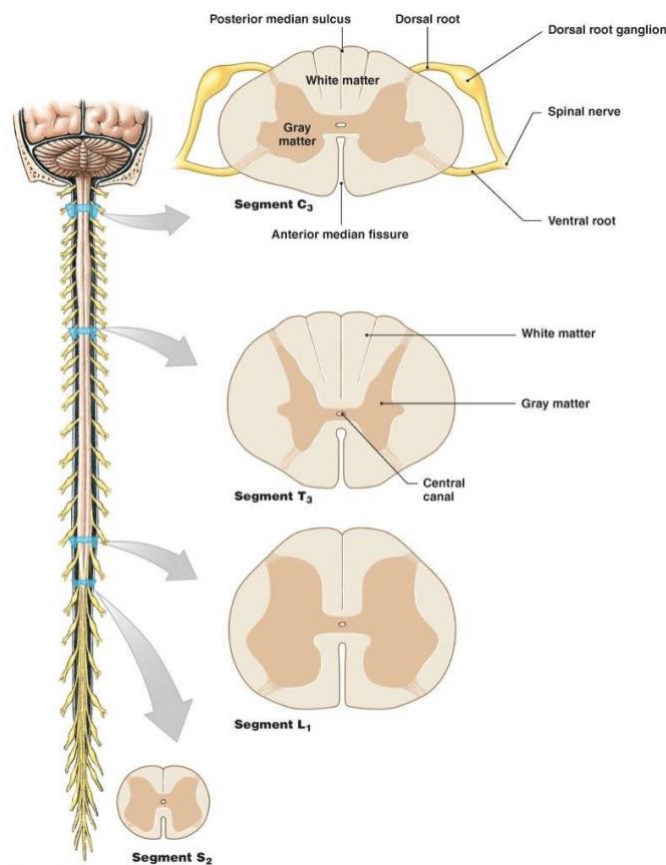


Figure 1: Spinal cord anatomy at representative segmental levels (cervical, thoracic, lumbar, sacral). Reproduced from [5].

Another distinction for later discussion of data processing is that, compared with the brain's highly directional and complex neural pathways, the myelin-coated nerve fibers run primarily unidirectionally along the cord [6].

The spinal cord's average length is between 43 and 45 cm, and its cross-section ranges from 51.5 to 96.9 mm² [7]. Compared to the brain's substantially larger, egg-shaped geometry, the cord's size makes it difficult to obtain high-resolution images of its microstructures.

Finally, another challenge in spinal cord imaging is the surrounding geometry. Bone, air-tissue interfaces near the lungs, and breathing and heartbeats all affect signal quality when scanning the cord, whereas these issues are less pronounced in the brain due to its greater distance from other organs [8].

1.2 Quantitative Susceptibility Mapping

QSM relies on a specific MRI image sequence to produce meaningful maps. Called a multi-echo gradient-recalled echo (meGRE) sequence, this acquisition protocol produces magnitude and phase images that map tissue T2* decay time and phase across each echo pulse, thereby providing the phase evolution over time, as further discussed in Section 2.2. QSM uses the phase map from this acquisition to build an image showing the tissue magnetic susceptibility through three key steps: phase unwrapping, background field removal, and dipole inversion, to be discussed in Section 3. The magnitude is important for creating a mask or mapping regions of interest in the cord.

The final output of QSM is called a χ -map (chi-map), which represents the bulk magnetic susceptibility of the tissue at each voxel. This image is a quantitative representation of

the tissue's paramagnetic or diamagnetic state at a given point, relative to a known reference, such as cerebrospinal fluid, or to the tissue's total average susceptibility.

The main contributors to overall susceptibility in WM and GM, as previously mentioned, are iron (paramagnetic) and myelin (diamagnetic) [1]. Because QSM provides the bulk susceptibility of the tissue at each point, susceptibility source separation is a further method built upon QSM to split this bulk χ -map into its χ -positive and χ -negative components, providing higher-detail information about the tissue. In a bulk χ -map, positive and negative effects may cancel each other within a voxel [3]. The extent of diamagnetism and paramagnetism exhibited by each chemical has been validated against post-mortem tissue composition across multiple studies [9], [10]. One of the susceptibility source separation methods discussed in this report, ANISO-QSM (Anisotropic Nonlinear Inversion for Susceptibility-separation Optimization), uses additional information on spinal cord geometry and fiber orientation to further improve the separation of iron and myelin [11].

1.3 Disease Identification through Quantitative Imaging

QSM is a valuable area of research in the MRI community because changes in iron and myelin are directly related to neurodegenerative diseases, especially multiple sclerosis. Paramagnetic rim lesions are a specific marker seen in MS patients; they are rings of iron-filled immune cells that appear around chronic active brain lesions [12]. Demyelination is another change seen in MS, the loss or damage of the myelin sheath that protects neurons and improves the conduction of electrical signals throughout the body [2].

QSM enables direct tracking of myelin loss or iron accumulation, rather than relying on visual inspection of qualitative images, making this method crucial for understanding these neurodegenerative diseases.

While QSM and related methods are far more extensively validated in the brain, the spinal cord, another important part of MS diagnosis, is largely untested using these methods, due to the physical constraints of the cord, and unoptimized algorithms suited for the brain [8]. The development of these pipelines on spinal cord datasets enables validation and correlation of QSM and susceptibility source separation against ground truth. This will provide momentum for the further adoption of QSM in the cord, with the ultimate goal of enabling reliable in vivo source separation of susceptibility for monitoring iron and myelin changes in spinal cord MS patients.

1.4 Purpose and Scope

This project was proposed by Dr. Eva Alonso Ortiz, Director at the NeuroPoly Lab, Polytechnique Montréal, as part of the author's co-op work term. The project's purpose was to develop a QSM processing pipeline suited to a 2D ex vivo spinal cord dataset, apply Chi-separation and ANISO-QSM susceptibility source separation methods, and assess the accuracy of these outputs against a histology-derived ground truth. This ex vivo validation is intended to support one of the lab's overarching interests: validating in vivo QSM and susceptibility source separation for in vivo spinal cord imaging.

The scope of this work was limited to a single ex vivo spinal cord sample, acquired as a 2D slice rather than a 3D volume, constraining the QSM and susceptibility source-separation pipelines to single-slice processing. No diffusion-weighted imaging (DWI) was acquired, so the fiber-orientation information required by ANISO-QSM was assumed to

be population-average values from the literature rather than measured directly. The available histology was limited to a myelin-specific stain, so validation against ground truth was restricted to the myelin-sensitive (χ -negative) outputs of each method; no iron-sensitive stain was available to validate the χ -positive (iron) outputs.

2 Data Description and Acquisition Protocols

The data used in this study consist of MRI acquisitions and a corresponding myelin-stained histological image from an ex vivo human spinal cord sample. Choosing a dataset that has been scanned with multiple protocols and including a ground truth in the histology data to validate any quantitative processing is key to providing evidence that the quantitative methods tested are accurate relative to the true amount of myelin determined from the microscopy image.

2.1 Overview of Ex Vivo Spinal Cord Data with Corresponding Histology

The data, specifically in 2D, are of the spinal cord at the T6 vertebral level, acquired and scanned by the NeuroPoly Lab in 2017 [13]. The sample was fixed in 4% paraformaldehyde (PFA) that was buffered (dissolved) in phosphate-buffered saline (PBS). PFA prevents decomposition, allowing for handling and scanning, and when dissolved in PBS, it prevents tissue swelling or shrinkage [14].

After the fixation period, there is a PBS wash to remove any of the unbound PFA, which prevents continued alteration of the tissue's susceptibility properties. This takes place over several days, and the tissue is finally stored in PBS to keep it hydrated.

The tissue was then osmified in 2% osmium tetroxide (OsO_4), which stains the myelin sheaths dark black. The sample was then embedded in resin and scanned using electron microscopy, which highlights the heavy-metal-stained myelin. With this microscopy image, the myelin volume fraction (MVF) was calculated by the NeuroPoly Lab and then mapped into MRI space for future use. This tissue preparation and histology workflow was adapted from the protocol established for ex vivo imaging of dog spinal cord [15].

2.2 MRI Acquisition Protocols

T2 is the intrinsic transverse relaxation time of a tissue, reflecting molecular-level interactions between spins. T2* is the effective transverse relaxation time observed in gradient echo acquisitions. T2* is always shorter than or equal to T2, as it additionally reflects the signal decay from local magnetic field inhomogeneities, which includes those caused by tissue magnetic susceptibility.

Both T2 and T2* are sensitive to a tissue's composition (i.e., its myelin and iron content), making them fundamental starting points for quantitative, noninvasive characterization of tissue composition and identification of pathological changes [1], [16].

The meGRE sequence is a gradient-echo acquisition without a refocusing pulse, sampled at multiple echo times. Because the resulting signal decay reflects both intrinsic tissue relaxation and additional reversible dephasing due to local-field inhomogeneities, the meGRE magnitude decay across echoes is used to map T2* [17].

The multi-echo spin-echo (MESE) sequence, by contrast, uses a series of 180° refocusing pulses that cancel this reversible dephasing at each echo. The resulting MESE magnitude decay isolates the tissue's true transverse relaxation and is used to map T2 [17].

All sequences used a field of view (FOV) of $10 \times 10 \times 3 \text{ mm}^3$ and a spatial resolution of $0.15 \times 0.15 \times 3 \text{ mm}^3$. The meGRE sequence was acquired with a repetition time (TR) of 300 ms, an echo time (TE) series of $TE1/\Delta TE/TE8 = 2.79/2.79/22.32 \text{ ms}$ (8 echoes), and a flip angle of $\alpha = 40^\circ$. The MESE sequence was acquired with a TR of 3000 ms and a TE series of $TE1/\Delta TE/TE32 = 10/10/320 \text{ ms}$ (32 echoes) [13].

3 Pipeline Development and Methods

A multitude of MRI processing toolboxes exist for quantitative imaging, but most consist of methods optimized and studied in the brain. This section describes the toolboxes used and how each was applied to do QSM, susceptibility source separation, and MWF mapping on the 2D spinal cord data.

3.1 Software Tools

The software tools used for scientific MRI image processing consist of open-source toolboxes, mainly written in MATLAB and Python. Typical toolboxes offer the option to use a graphical user interface (GUI) or to call functions directly from the command line.

3.1.1 *SEPIA Toolbox*

The SEPIA (Susceptibility mapping Pipeline tool for phase images) Toolbox [18] serves as a GUI that enables users to interface with multiple MATLAB QSM toolboxes, such as STI-Suite (Susceptibility Tensor Imaging Suite) [19] (used for QSM in this report), to create custom processing pipelines tailored to the acquired data. SEPIA includes all the main QSM processing steps, from unwrapping to dipole inversion, and provides a selection of methods available for each step. While optimized for brain and 3D volume processing, SEPIA is the starting point for developing a 2D ex vivo processing pipeline. SEPIA is also used to perform R_2 and R_2^* mapping (the transverse and effective transverse relaxation rates, respectively, defined further in Section 3.3.1), a required input for susceptibility source separation.

3.1.2 *Susceptibility Source Separation Toolboxes*

The primary toolbox for susceptibility source separation is the Chi-separation toolbox [3]. This toolbox performs isotropic susceptibility source separation, again optimized for the

brain, and requires the local field as input, which is an intermediate step in QSM processing. The toolbox being compared to the Chi-separation toolbox is called ANISO-QSM [11], a new method that uses myelin fiber orientation as additional input in its processing algorithm.

3.1.3 Multicomponent-T2-Toolbox

The Multicomponent-T2-Toolbox is used to process MESE acquisition to produce an MWF map [20]. MWF is a well-validated quantitative technique [21] that will serve as an additional mode of comparison and validation.

3.1.4 Spinal Cord Toolbox and ANTs Co-registration

Spinal Cord Toolbox (SCT) [22] and Advanced Normalization Tools (ANTs) [23] are both toolboxes more tailored to orientation and data handling for spinal cord images. SCT provides segmentation and masking of spinal cord images, and also allows header manipulation for co-registration. ANTs is specifically for image orientation and is used to warp images with different resolutions or FOVs into a common space for analysis.

3.1.5 FSLeaves and ITK-SNAP Image Viewers

Because MRI images are 3D and sometimes 4D (x, y, z, time), specialized image viewers are required to review the output at different processing steps and to manually map masks onto the image for co-registration and comparison. FSLeaves [24] and ITK-SNAP [25] are both image viewers designed specifically for MRI and provide intuitive interfaces for viewing processed data.

3.2 QSM Pipeline

As described in Section 2.2, the initial QSM processing pipeline uses the magnitude and phase from the meGRE acquisition. The first step in the process is the unwrapping of the phase that has evolved over time due to tissue inhomogeneities. The phase offset

accumulates across multiple echoes, allowing one to map local phase changes due to tissue composition. Because phase is an angle and can only be measured from 0 to 2π , the initial phase image shows discontinuities that must be corrected, since the true phase evolution is gradual over time. Phase unwrapping is a technique that detects those phase jumps and recovers the true phase.

The technique chosen for unwrapping the phase of this single slice ex vivo data was Laplacian Phase Unwrapping from the STI-Suite toolbox. This toolbox provides a 2D method developed to work with single-slice data. The phase was unwrapped for each individual echo, and the echoes were then fitted using the magnitude data as weighting to produce a total-field map representing all sources of inhomogeneity in the scan (i.e., scanner hardware and local tissue fields), as shown in Figure 2.

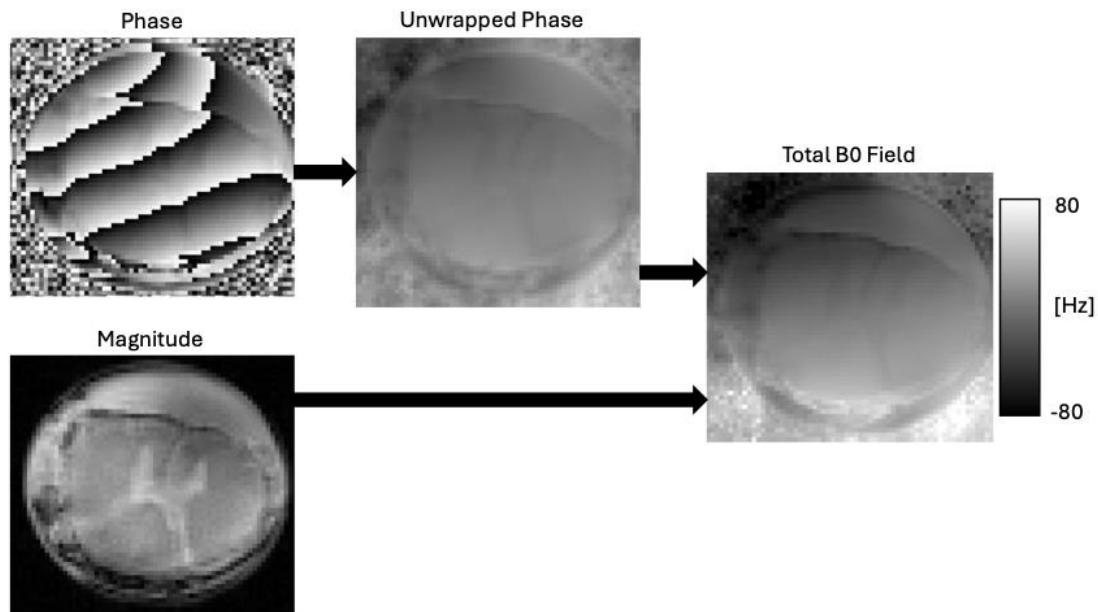


Figure 2: The initial QSM processing steps, the raw phase and magnitude from the acquisition are used to create the total field relative to the main static magnetic field (B_0), once the phase has been unwrapped.

Now with the total field map, the next step is to isolate the inhomogeneities caused by tissue, which reflect the tissue's magnetic susceptibility. The background field consists of large-scale contributions from the scanner and external sources that must be separated from the fine, sharp variations caused by the tissue's distribution.

A custom Gaussian high-pass filter (HPF) was initially chosen for background field removal because its implementation required fewer assumptions for the single 2D slice geometry of this dataset, unlike many established methods (such as SHARP-based approaches) that rely on a spherical convolution kernel operating over a 3D volume. The idea was that convolving the field map with a Gaussian kernel would blur out and isolate the smooth, large, spatially varying background field, so subtracting that blurred estimate from the original field map would leave behind the sharp, local field of interest. In practice, this approach produced major edge artifacts in the resulting local field, as the smoothing kernel blurred across the sharp tissue boundary rather than respecting it.

Based on further review of existing methods, 2D V-SHARP (Variable-kernel Sophisticated Harmonic Artifact Reduction for Phase data) from the STI-Suite toolbox was used instead; it varies its filtering radius near the tissue boundary to better preserve the edges, unlike a fixed Gaussian kernel, thereby reducing the edge artifacts seen with the initial approach. The standard V-SHARP kernel is a spherical mean value (SMV) kernel that requires a 3D volume of neighboring slices to compute. The 2D implementation instead uses a circular kernel confined to the plane of the single acquired slice, which is what allows the same variable-radius, boundary-preserving approach to be applied to this single-slice dataset without requiring volumetric support.

To optimize the 2D V-SHARP algorithm for this specific dataset, a manual test of different kernel sizes was conducted, using the contrast-to-noise ratio (CNR) between white and grey matter as the objective metric to produce the optimal local field output (Figures 3 and 4; results discussed in Section 4.1).

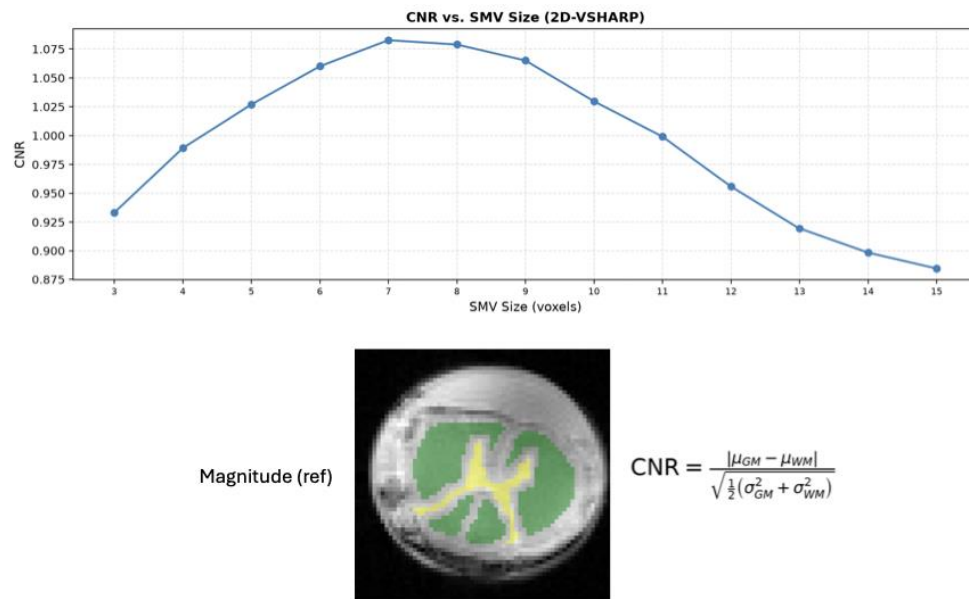


Figure 3: Contrast-to-noise analysis between GM and WM with different SMV kernel sizes to determine the optimal settings for 2D V-SHARP.

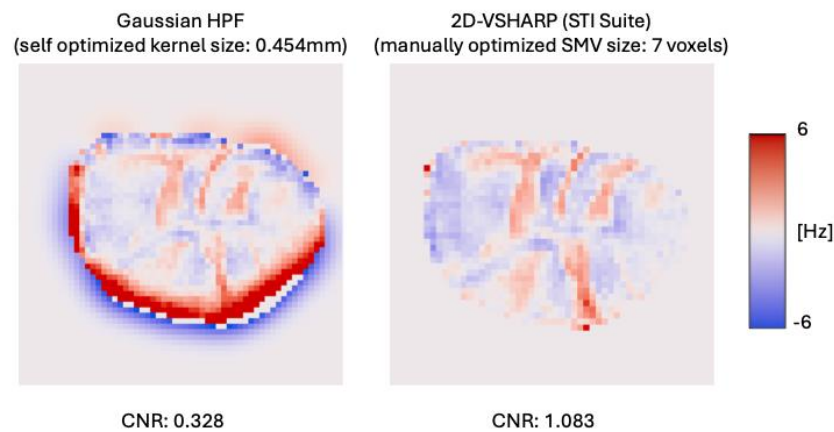


Figure 4: Background field result from Gaussian HPF implementation (left) and 2D V-SHARP (right). The edge artifacts in the Gaussian HPF image reduce CNR compared with 2D V-SHARP.

With the local field isolated, the final step in the QSM pipeline is dipole inversion, which converts the local field map into a quantitative susceptibility χ -map. This step inverts the physical relationship between a tissue's magnetic susceptibility and the field it produces, which is mathematically described as a convolution with a unit-dipole kernel [26], [27]. This dipole kernel, seen in Figure 5, in the spatial frequency domain, has a conical surface of near-zero values at the "magic angle" (approximately 54.7° relative to B_0), where the field-to-susceptibility relationship becomes unstable; division by these near-zero values amplifies noise and produces streaking artifacts seen in unregularized QSM reconstructions [28]. Dipole inversion is inherently ill-posed, and a range of methods exists to stabilize the inversion and minimize these artifacts.

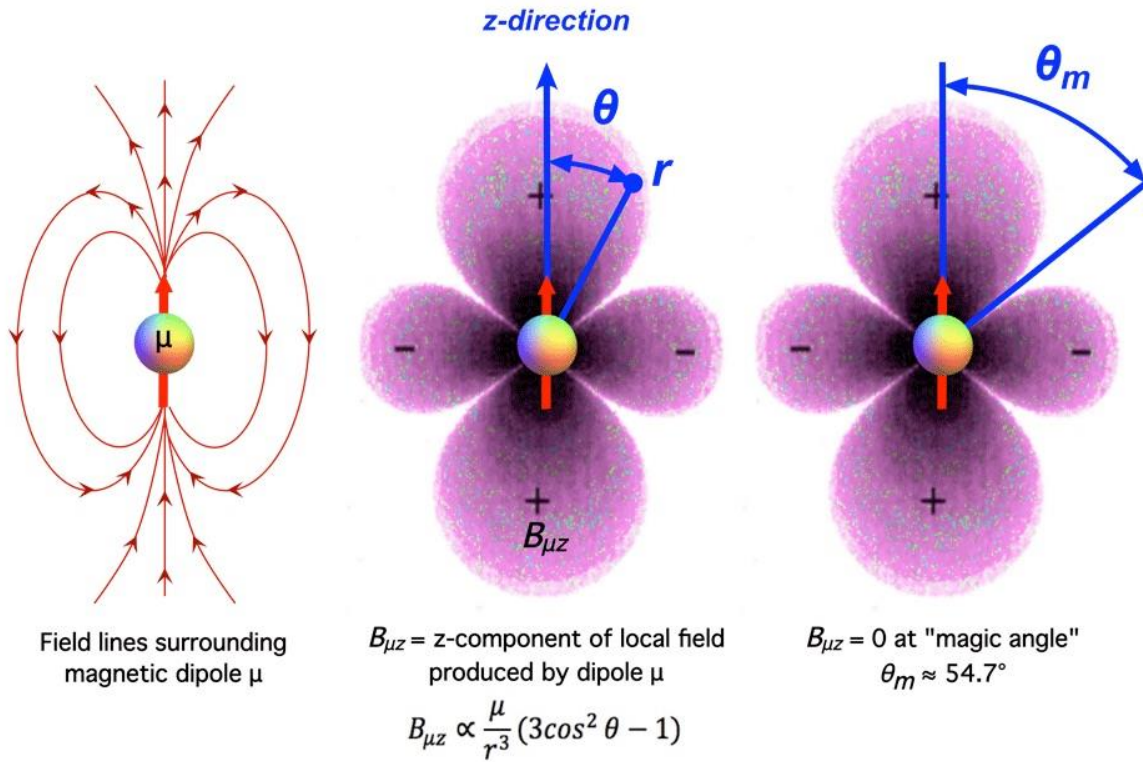


Figure 5: The unit dipole. Reproduced from [29].

Three dipole inversion methods were compared for this dataset: Thresholded k-space Division (TKD) [28], Closed-form L2-regularized inversion [30], and Morphology Enabled Dipole Inversion (MEDI) [31]. These methods were provided in the SEPIA toolbox, but function wrappers were built and used for this specific QSM pipeline.

TKD stabilizes the inversion by excluding the near-zero region of the dipole kernel with a threshold before dividing, at the cost of accuracy near that threshold boundary. The closed-form method instead applies Tikhonov-style L2 regularization weighting, while MEDI iteratively refines the solution using edge information from the magnitude image as an anatomical prior, suppressing streaking while keeping structural detail. Both the closed-form and MEDI implementations self-optimize regularization strength, whereas TKD requires manual threshold selection.

Using the same CNR-based approach as before for the background field removal kernel size, the TKD threshold was optimized by sweeping over candidate threshold values and selecting the one that maximized the contrast-to-noise ratio between white and grey matter (Figures 6 and 7; results discussed in Section 4.1).

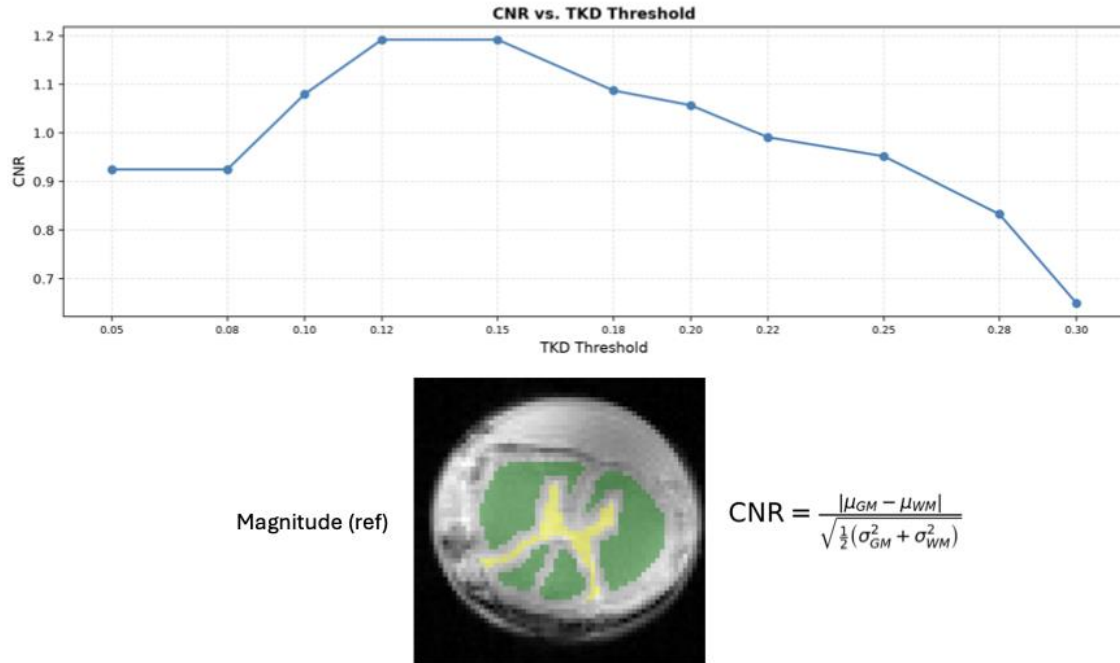


Figure 6: CNR analysis between WM and GM with varying TKD threshold levels.

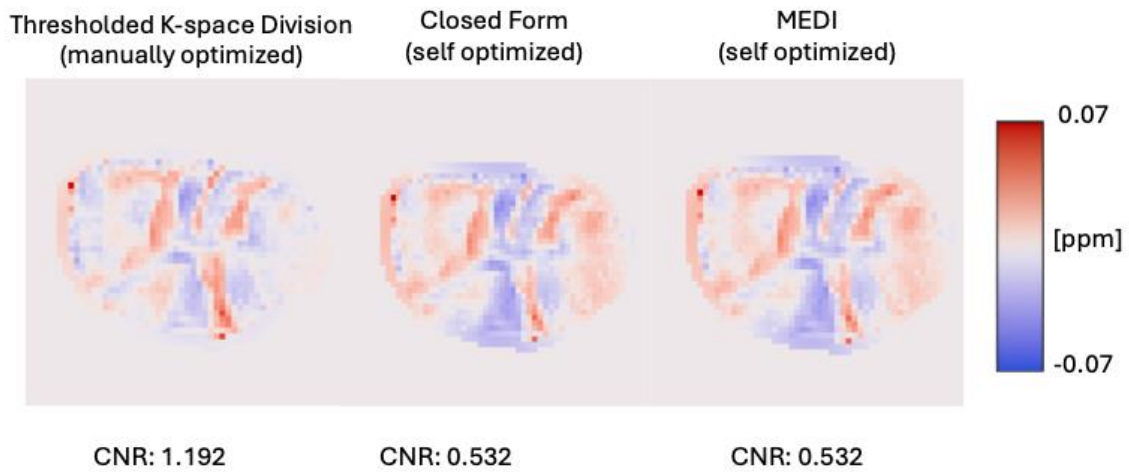


Figure 7: Three dipole inversion methods, including the CNR between WM and GM of each method. TKD (left), Closed Form (middle), MEDI (right).

3.3 Susceptibility Source Separation Pipeline

Susceptibility source separation refers to a family of methods built upon the QSM pipeline that separate the bulk magnetic susceptibility into paramagnetic (iron) and diamagnetic (myelin) components, addressing the limitation that a single bulk susceptibility value cannot distinguish between two sources of opposite sign within a voxel. Two susceptibility source separation methods are used in this report: Chi-separation, described in Section 3.3.1, and ANISO-QSM, an anisotropic method described in Section 3.3.2.

3.3.1 Chi-separation

Chi-separation [3] requires the magnitude image from the meGRE acquisition, the local field obtained as an intermediate output of the QSM pipeline described in Section 3.2, and an $R2'$ ($R2$ -prime) map. The local field is signed and reflects the difference between paramagnetic and diamagnetic contributions, whereas $R2'$ reflects their combined effect, since both sources contribute to signal dephasing regardless of sign. Combining the two provides the information needed to solve for each component separately.

To obtain the $R2'$ map, $R2^*$ and $R2$ were first computed from the meGRE and MESE magnitude data, respectively, by fitting the signal decay across echoes to a mono-exponential decay model using a non-linear least-squares method in SEPIA. $R2^*$ and $R2$ are the inverse of the previously described $T2^*$ and $T2$ relaxation times ($R2^* = 1/T2^*$, $R2 = 1/T2$) and represent the corresponding relaxation rates, which are how quickly, rather than how long, the signals decay. The $R2'$ map was then calculated as the difference between the two ($R2' = R2^* - R2$), isolating the reversible relaxation component associated with the susceptibility sources of interest and providing a measure of the total,

unsigned concentration of paramagnetic and diamagnetic sources. The $R2'$ pipeline is shown in Figure 8.

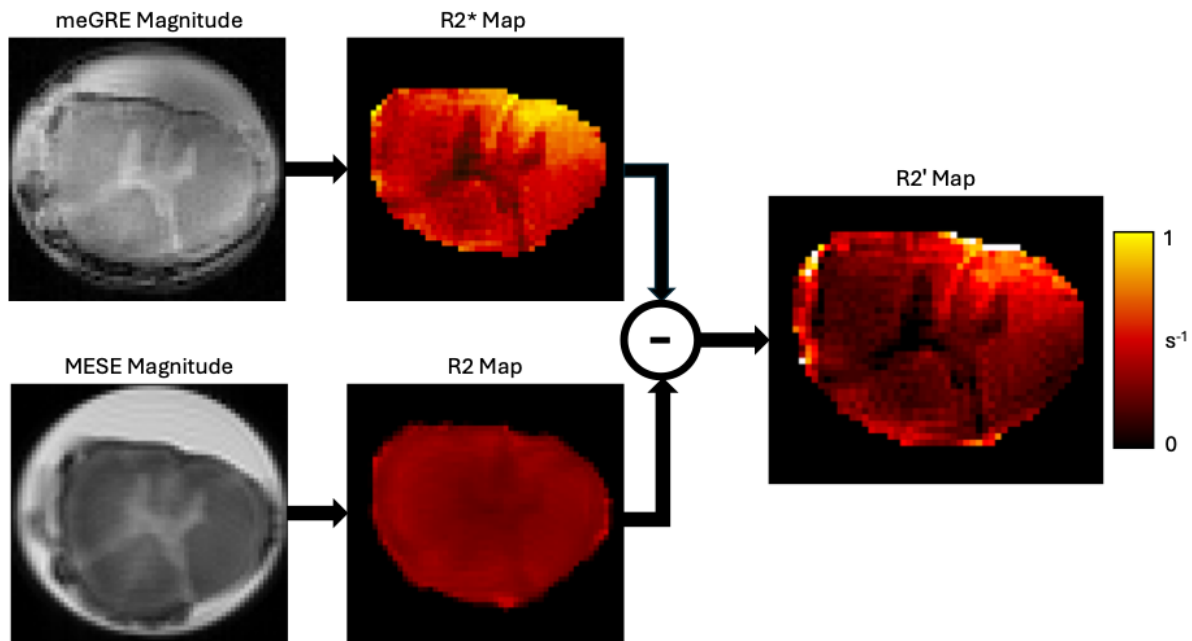


Figure 8: Pipeline for creation of $R2'$ map using the meGRE and MESE acquisitions.

Together, the magnitude image, local field, and $R2'$ map enable the Chi-separation algorithm to compute the individual paramagnetic and diamagnetic susceptibility maps, even when both sources coexist within the same voxel.

The resulting χ -positive and χ -negative maps produced by Chi-separation are shown in Figure 9; the qualitative results are discussed in Section 4.2.

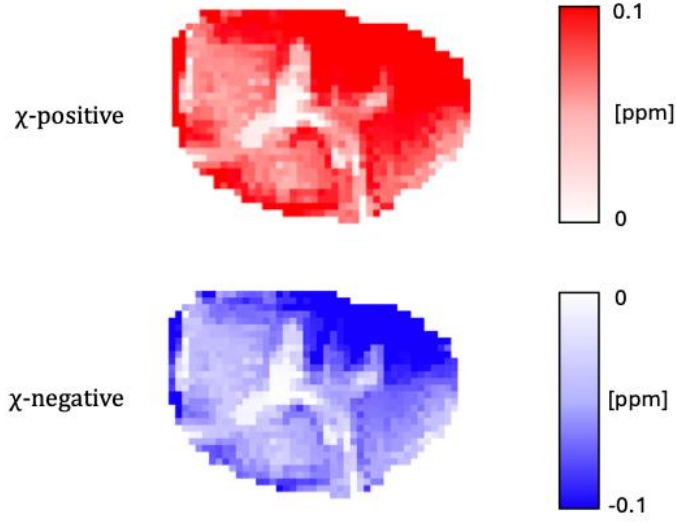


Figure 9: χ -negative and χ -positive maps created using the Chi-separation toolbox.

3.3.2 ANISO-QSM

ANISO-QSM extends susceptibility source separation by incorporating information on myelin fiber orientation and anisotropy to improve the accuracy of susceptibility calculation in WM.

ANISO-QSM requires a DWI acquisition to obtain information about the tissue's anisotropies. Because no DWI was acquired for this sample when it was scanned in 2017, research was conducted to provide the algorithm with the best assumptions for the needed fractional anisotropy (FA) and fiber angle maps. It was found that FA in the spinal cord is high [32], indicating that the fibers have little overlap. The average value from the Spine Generic dataset, a large open-source dataset spanning 260 participants, was approximately 0.7. For the fiber angle map, which is the angle of the tissue relative to the scanner B0 field, it was assumed to be 0° as the ex vivo sample was positioned parallel

to the field during scanning. The resulting output maps are shown in Figure 10; the qualitative results are discussed in Section 4.2.

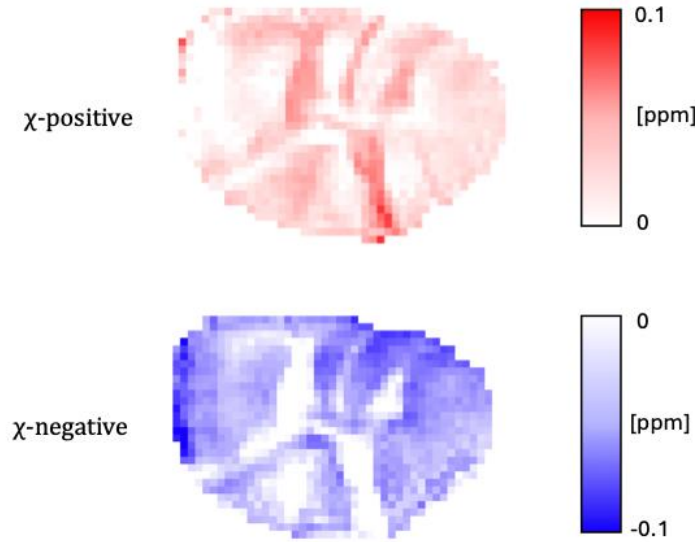


Figure 10: ANISO-QSM χ -positive and χ -negative maps.

3.4 Myelin Water Fraction Mapping

MWF mapping, another myelin-sensitive quantitative technique, was also applied to the dataset to provide an additional way to compare and validate the susceptibility source separation methods.

The MESE acquisition was processed with the Multicomponent-T2-Toolbox (Section 3.1.3) to generate MWF maps. The toolbox fits the multi-echo signal decay in each voxel to a distribution of T2 components, separating a short-T2 myelin water pool from the longer-T2 intra- and extracellular water pool; the myelin water fraction is then calculated as the proportion of signal that comes from the myelin water pool to the entire water pool

in the specific area of tissue. The toolbox's default cutoff of 40 ms is an assumption established for in vivo imaging [33], but because this dataset is ex vivo, fixation and prolonged PBS storage are known to alter tissue T2 relaxation times [34], meaning this 40 ms assumption may not accurately separate the myelin and non-myelin water pools for this sample.

To account for this, an initial MWF map was first generated using the standard 40 ms cutoff, and Otsu's thresholding method [35], implemented in Python, was applied to the resulting T2 distribution to identify a more specific cutoff between the myelin and non-myelin water pools, determined to be 29.3 ms as seen in Figure 11. This cutoff was then used to recompute the final MWF map shown in Figure 12.

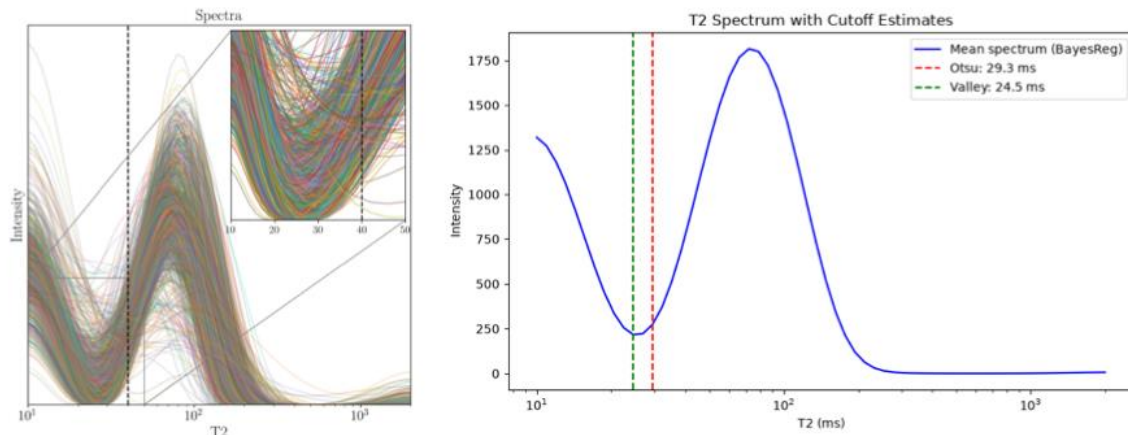


Figure 11: The initial T2 spectrum produced using a 40 ms T2 cutoff. It can be seen visually that the true cutoff is before 40 ms, which is then verified on the right plot, where Otsu's method determined a cutoff of 29.3 ms.

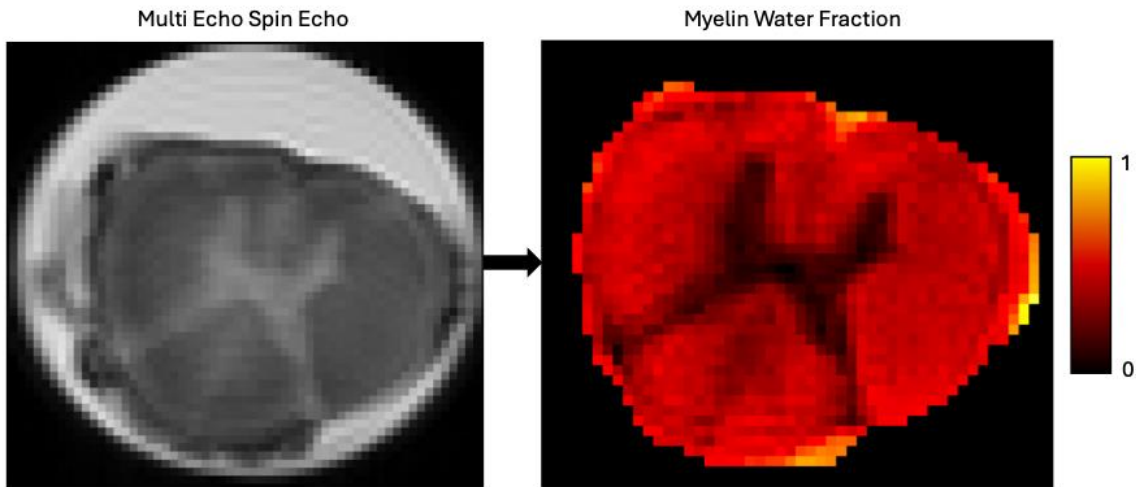


Figure 12: The final MWF map, created from the MESE acquisition, with the T2 cutoff at 29.3 ms.

3.5 Co-registration

The histology image and MWF image come from different references than the susceptibility source separation methods, so the anatomical structures are not automatically aligned and cannot be directly compared. Before comparison, these images have to be co-registered to the same reference space, in this case, the meGRE magnitude from which both susceptibility separation outputs originate.

Co-registration allows images with different FOVs or resolutions to be warped to match the reference, which will align the WM and GM to enable specific regions to be compared. If done perfectly, each voxel corresponds to the same tissue area across images.

While SCT provides methods for co-registering 3D spinal cord acquisitions, a custom pipeline was developed to reliably and repeatably map 2D images to one another. This pipeline uses SCT to read the image header and obtain orientation and resolution data. Next, the reference orientation is applied to the input image, and the user can view it to

determine whether any manual rotations or reflections are needed. In the next step, once any manual adjustments are applied, ANTs is used to warp the image to the reference using a mask. A WM mask (the highlighted green area in the magnitude image in Figure 13) was created for both the input and reference images, enabling the ANTs algorithm to better align WM and GM structures and improve warping.

Figure 13 shows the before-and-after co-registration of the MVF and MWF maps to the magnitude reference.

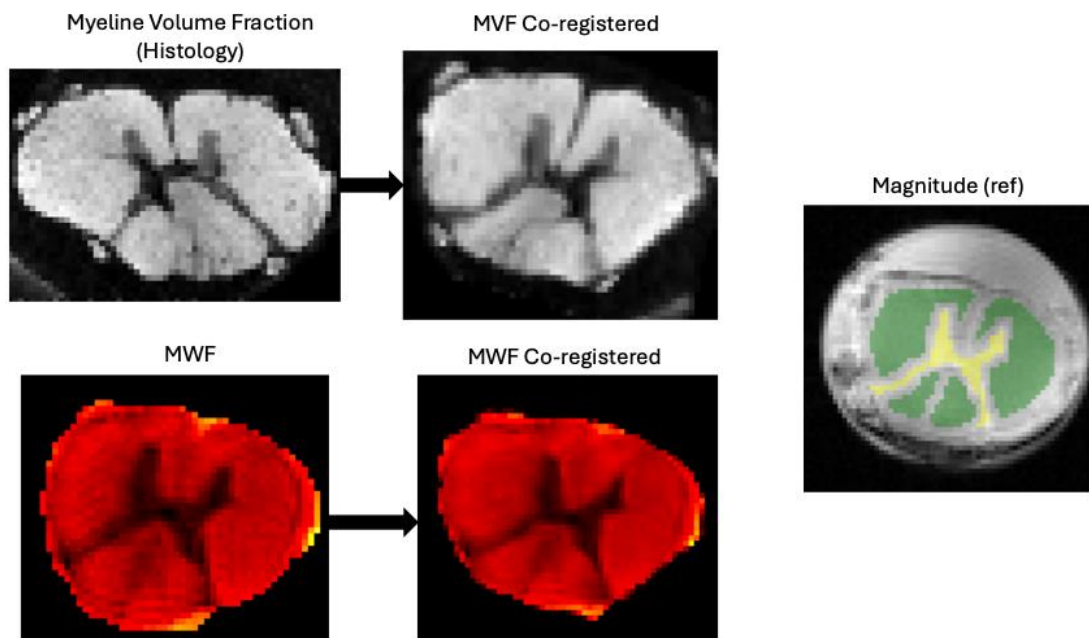


Figure 13: MVF and MWF maps, both coregistered to the meGRE magnitude, with WM and GM aligned.

4 Results

While a promising approach for QSM was described in this report, it cannot be validated directly against the histological ground truth because the histology is myelin-stained and QSM represents both myelin and iron susceptibility in a single map. The χ -negative maps from susceptibility source separation, therefore, provide the most direct means of evaluating the background field removal techniques described in Section 3.

The Chi-separation and ANISO-QSM χ -negative maps, MVF, and MWF were compared because they are all myelin-sensitive and allow direct comparison of myelin accuracy against the MVF ground truth.

All comparisons in this section use the Pearson correlation coefficient, a measure of the strength and direction of the linear relationship between two variables. Its value ranges from -1 to 1: a value near 1 indicates a strong positive linear relationship, near -1 indicates a strong negative linear relationship, and near 0 indicates little to no linear relationship between the two maps being compared. All correlations were assessed using a two-tailed test with a significance threshold of $\alpha = 0.05$.

4.1 Pipeline Optimization Results

For background field removal, a kernel size of 7 voxels produced the highest CNR (1.083) between white and grey matter using 2D V-SHARP (Figure 3). The CNR achieved with the Gaussian high-pass filter implementation was far below that of 2D V-SHARP (0.328 vs. 1.083; see Figure 4).

For dipole inversion, the TKD threshold sweep identified an optimal threshold of 0.15, which maximized the CNR between white and grey matter (1.192; see Figure 6). Comparing all three dipole inversion methods directly, TKD produced a substantially

higher CNR and clearer contrast between WM and GM than either the closed-form or MEDI reconstructions (1.192 vs. 0.532 vs. 0.532; see Figure 7).

4.2 Susceptibility Source Separation Outputs

The Chi-separation method was unable to resolve the positive and negative components of the tissue in this 2D dataset; the χ -positive map (Figure 9, Section 3.3.1) visually mirrored the χ -negative map, showing high positive susceptibility in the WM and low positive susceptibility in the GM, when the opposite is expected.

The output maps produced by ANISO-QSM (Figure 10, Section 3.3.2) show clear WM and GM contrast in both the positive and negative susceptibility maps.

4.3 Voxel-wise Comparison

For the voxel-wise comparison, Pearson correlation coefficients were computed on a voxel-by-voxel basis across the full cord mask, comparing each pair of the following maps: the Chi-separation χ -negative map, the ANISO-QSM χ -negative map, MWF, and MVF (the histology-derived ground truth). Every voxel in the mask was included, without restriction to specific anatomical regions, yielding sample sizes ranging from 828 to 1068 voxels depending on the map pair compared.

The resulting correlation matrix is shown in Figure 14. MWF and MVF showed a Pearson correlation of 0.87. The Chi-separation χ -negative map correlated with the ANISO-QSM χ -negative map at 0.57, with MWF at 0.33, and with MVF at 0.26. The ANISO-QSM χ -negative map correlated with MWF at 0.21 and with MVF at 0.14. All reported correlations were statistically significant ($p < 0.05$).

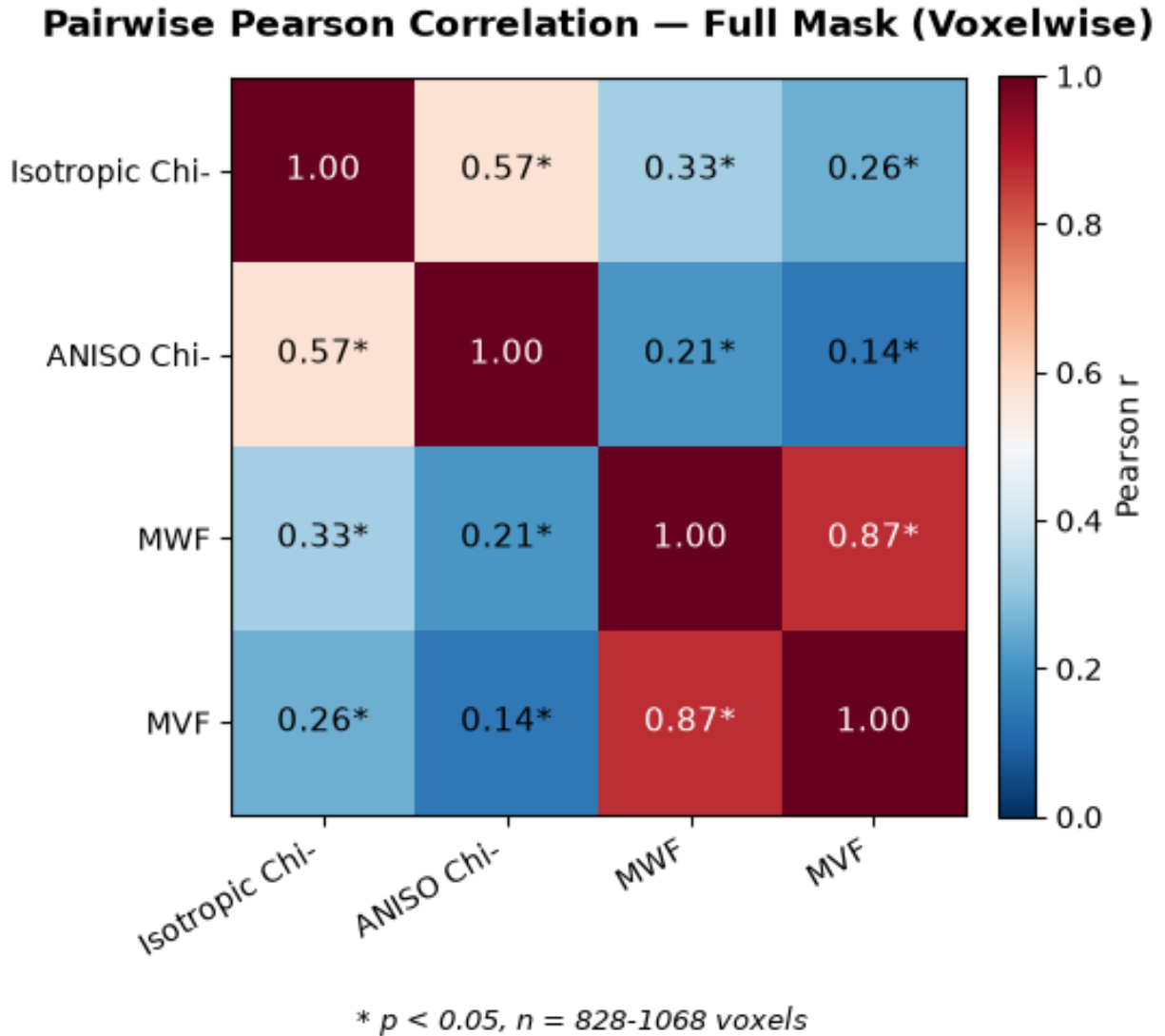


Figure 14: Pearson correlation matrix with full mask voxel-wise comparison results between methods.

4.4 Region of Interest Comparison

For the region-of-interest (ROI) comparison, a mask was created that includes 14 white matter tract regions, defined using a histological image at the T6 vertebral level with labeled regions [36], seen in Figure 15. The mask was applied to each map and the mean value for each region was extracted, and Pearson correlation coefficients were computed across these 14 tract-wise means, rather than on an individual-voxel basis.

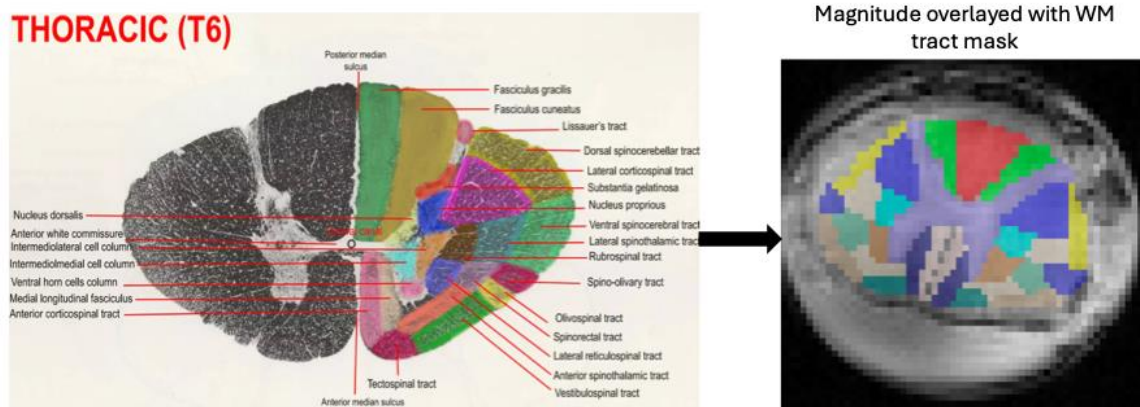
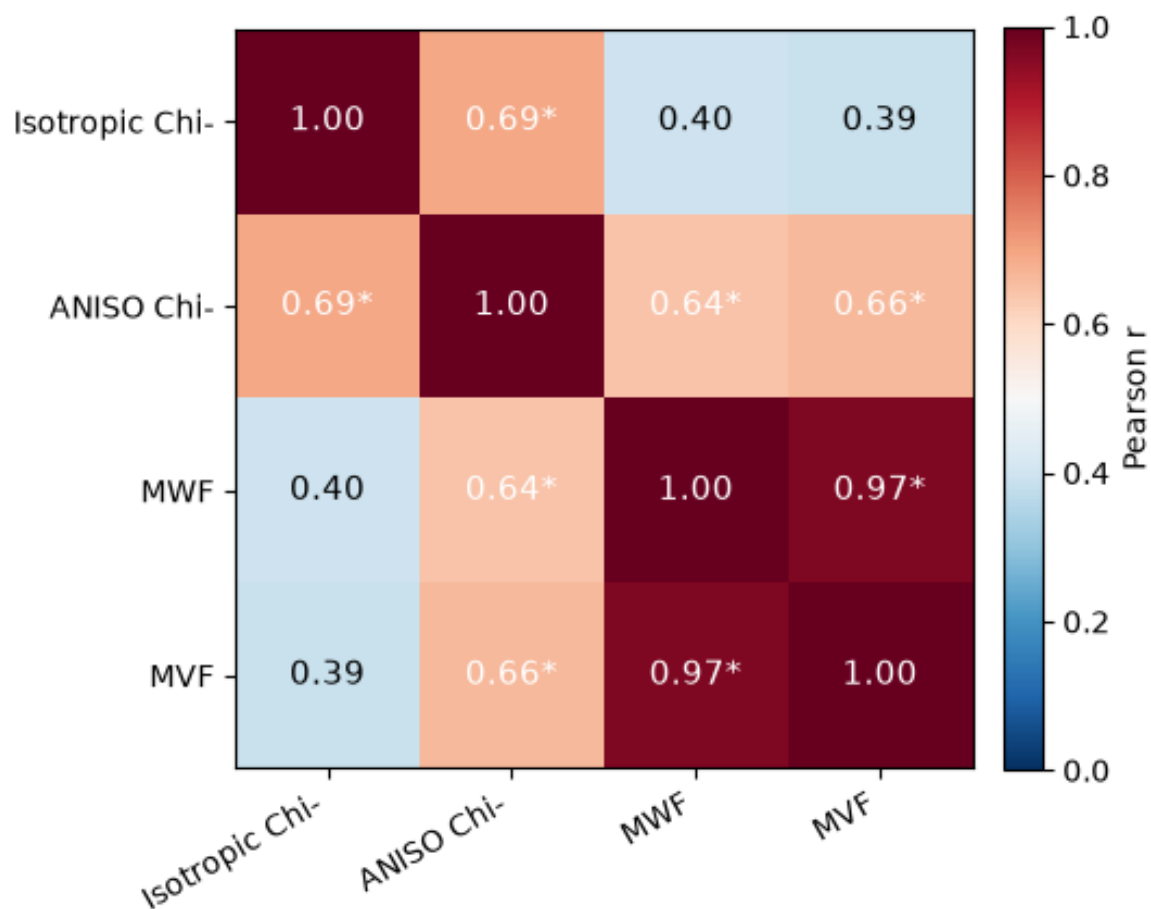


Figure 15: A manual WM segmentation, based on a tract-labeled histology image at the T6 vertebral level, reproduced from [36].

The resulting correlation matrix is shown in Figure 16. MWF and MVF showed a Pearson correlation of 0.97. The Chi-separation χ -negative map correlated with the ANISO-QSM χ -negative map at 0.69. The ANISO-QSM χ -negative map correlated with MWF at 0.64 and with MVF at 0.66. The Chi-separation χ -negative map correlated with MWF at 0.40 and with MVF at 0.39; these two correlations were not statistically significant ($p \geq 0.05$). All other reported correlations were statistically significant ($p < 0.05$).

Pairwise Pearson Correlation — WM Tract Mask (ROI-wise)



* $p < 0.05$, $n = 14$ ROIs

Figure 16: ROI-wise Pearson correlation matrix between the different techniques.

5 Discussion

This work developed and applied a 2D QSM pipeline to an ex vivo human spinal cord sample and then used its local-field output to process two susceptibility-source separation methods (Chi-separation and ANISO-QSM). The maps from these methods were compared against each other, as well as an MWF map and a histology-derived MVF as ground truth. The results provide an early data point toward validating these methods in the cord, an area that remains comparatively underexplored relative to their established use in the brain [8].

The strong, statistically significant correlation between MWF and MVF at the ROI level ($r = 0.97$) supports the validity of the ex vivo MWF processing approach used in this work, including the Otsu-refined T2 cutoff used to account for fixation effects on tissue relaxation times. This agreement is consistent with prior histological validation of MWF imaging [16], [21], and indicates that MWF and MVF can reasonably serve as a shared reference point for evaluating the Chi-separation and ANISO-QSM outputs.

Comparing the Chi-separation and ANISO-QSM χ -negative maps against this shared reference shows that the voxel-wise correlations are weak for both methods (0.33 and 0.26 for Chi-separation with MWF and MVF, respectively; 0.21 and 0.14 for ANISO-QSM). These voxel-wise results are unlikely to be representative of either method's true agreement with the underlying tissue composition. At the resolution of this acquisition, partial volume effects are present within voxels (GM and WM within a voxel), and comparing voxels is not representative of any discrete anatomical structure. Co-registration compounds this problem, since small misalignments at tissue boundaries have a much larger relative effect on a single voxel's value than on a tract-averaged value.

This is a more plausible explanation for the weak voxel-wise correlations than any genuine difference in accuracy between the two methods.

The ROI-level comparison, which averages within anatomically defined white matter tracts and is less sensitive to partial-volume and co-registration misalignment, is therefore a more meaningful basis for comparing the two methods. At this level, ANISO-QSM correlated substantially more strongly with MWF and MVF (0.64 and 0.66, respectively) than Chi-separation did (0.40 and 0.39, neither of which reached statistical significance), indicating an advantage of incorporating fiber-orientation information.

From a qualitative standpoint, the Chi-separation method was unable to reliably resolve the χ -positive and χ -negative components in this 2D dataset, with the χ -positive map visually mirroring the χ -negative map rather than showing the expected inverse WM/GM contrast. One potential explanation is that the algorithm overcorrects in response to the large gradient observed in the $R2'$ map, which is also visible in the meGRE magnitude, thereby losing the smaller, sharper susceptibility changes in the process. ANISO-QSM produced clearer, more anatomically expected WM/GM contrast in both maps, suggesting that incorporating fiber-orientation information partially mitigated this overcorrection. Because the spinal cord's white matter is composed of densely packed, longitudinally-oriented fiber tracts [6], anisotropic effects on local susceptibility contrast may be more pronounced and more consistently modeled here than in the more heterogeneously-oriented white matter of the brain. This potentially explains why the anisotropic assumptions used in ANISO-QSM improved agreement with the myelin ground truth at the ROI level.

Several limitations constrain these findings. This analysis is based on a single ex vivo sample, so the reported correlations cannot yet be assessed for reproducibility across subjects. The 2D acquisition protocol limited the Chi-separation and ANISO-QSM pipelines to a single-slice representation of the local field and $R2'$, which likely contributed to the sign-resolution and gradient-overcorrection issues mentioned in Section 4.2. Because no DWI acquisition was available for this sample, the fractional anisotropy and fiber angle inputs to ANISO-QSM were drawn from population-average values from the Spine Generic dataset [32] rather than measured directly, which may not fully capture this sample's true fiber geometry. Finally, ex vivo fixation and prolonged buffer storage are known to alter tissue relaxation properties [14], [34], so the relationship between these MRI-derived measures and the underlying tissue composition may differ from what is observed in vivo.

Because the motivating goal of this work is to eventually support in vivo susceptibility source separation in the spinal cord for clinical applications such as MS monitoring, the ROI-level agreement between ANISO-QSM and the myelin ground truth observed here is an encouraging early sign. However, it must be confirmed on 3D, in vivo acquisitions before this pipeline can be considered clinically translatable.

6 Conclusions

1. A 2D QSM pipeline was developed, with the local-field output from the background-field removal step used as input to two susceptibility source separation methods: Chi-separation and ANISO-QSM. Additionally, MWF maps were created for additional comparison.
2. MWF and histology-derived MVF correlated strongly at the ROI level ($r = 0.97$), supporting the ex vivo MWF processing approach used as a validation reference.
3. Voxel-wise correlations between the χ -negative maps and the myelin ground truth were weak across both source separation methods, most likely due to partial volume effects and imperfect co-registration rather than a true difference in accuracy.
4. At the ROI level, ANISO-QSM correlated more strongly with the myelin ground truth than Chi-separation and produced clearer WM/GM contrast, indicating a benefit from incorporating fiber-orientation information.
5. These results represent a preliminary step toward validating QSM-based susceptibility source separation against histological ground truth in the spinal cord.

7 Recommendations

1. Acquire dedicated DWI for future ex vivo cord samples to obtain subject-specific FA and fiber angle maps, replacing the population-average assumptions used for ANISO-QSM in this work, at the cost of an additional acquisition time of approximately 20–30 minutes per sample.
2. Use a 3D meGRE acquisition protocol, rather than the 2D protocol used here, to address the sign-resolution and $R2'$ gradient-overcorrection issues that limited the Chi-separation method's performance, at the cost of increased acquisition and processing time, and the need to re-optimize the background field removal and dipole inversion steps for 3D data.
3. Expand the dataset to additional ex vivo spinal cord samples to test whether the correlation patterns observed here, particularly the ROI-level advantage of ANISO-QSM over Chi-separation, are reproducible across subjects, at the cost of the additional tissue procurement, scanning, and processing time for each new sample.
4. Extend this pipeline to in vivo spinal cord acquisitions once ex vivo validation is established across multiple samples, in support of the lab's motivating goal of enabling in vivo susceptibility source separation for spinal cord MS monitoring, at the cost of the additional scan-time constraints and patient-motion considerations, and time to develop pipelines for full cord mapping.

List of References

- [1] J. H. Duyn and J. Schenck, "Contributions to magnetic susceptibility of brain tissue," *NMR in Biomedicine*, vol. 30, no. 4, e3546, 2017.
- [2] D. S. Reich, C. F. Lucchinetti, and P. A. Calabresi, "Multiple Sclerosis," *New England Journal of Medicine*, vol. 378, no. 2, pp. 169–180, 2018.
- [3] H.-G. Shin et al., " χ -separation: Magnetic susceptibility source separation toward iron and myelin mapping in the brain," *NeuroImage*, vol. 240, art. 118371, 2021.
- [4] G. David et al., "Traumatic and nontraumatic spinal cord injury: pathological insights from neuroimaging," *Nature Reviews Neurology*, vol. 15, no. 12, pp. 718–731, 2019.
- [5] F. H. Martini, W. C. Ober, and J. L. Nath, *Visual Anatomy & Physiology*, 1st ed. San Francisco, CA, USA: Benjamin Cummings, 2011.
- [6] J.-L. Cui, C.-Y. Wen, Y. Hu, K.-C. Mak, K.-H. H. Mak, and K. D.-K. Luk, "Orientation entropy analysis of diffusion tensor in healthy and myelopathic spinal cord," *NeuroImage*, vol. 58, no. 4, pp. 1028–1033, 2011.
- [7] S. Bédard and J. Cohen-Adad, "Automatic measure and normalization of spinal cord cross-sectional area using the pontomedullary junction," *Frontiers in Neuroimaging*, vol. 1, art. 1031253, 2022.
- [8] P. W. Stroman et al., "The current state-of-the-art of spinal cord imaging: Methods," *NeuroImage*, vol. 84, pp. 1070–1081, 2014.
- [9] C. Langkammer et al., "Quantitative susceptibility mapping (QSM) as a means to measure brain iron? A post mortem validation study," *NeuroImage*, vol. 62, no. 3, pp. 1593–1599, 2012.

- [10] C. Stüber et al., "Myelin and iron concentration in the human brain: A quantitative study of MRI contrast," *NeuroImage*, vol. 93, pt. 1, pp. 95–106, 2014.
- [11] D. Ridani, Y. Diouane, B. De Leener, and E. Alonso Ortiz, "ANISO-QSM: An Anisotropic Nonlinear Inversion for Susceptibility-separation Optimization algorithm," in *Proc. 2026 ISMRM-ISMRT Annual Meeting and Exhibition, Cape Town, South Africa, 2026*.
- [12] P. Maggi et al., "Paramagnetic Rim Lesions Are Specific to Multiple Sclerosis: An International Multicenter 3T MRI Study," *Annals of Neurology*, vol. 88, no. 5, pp. 1034–1042, 2020.
- [13] NeuroPoly Lab, Polytechnique Montréal, Montréal, QC, Canada. Ex Vivo Human Spinal Cord Dataset: meGRE and MESE Acquisition Parameters. (2017). Accessed: May 22, 2026. [Intranet].
- [14] C. H. Fox, F. B. Johnson, J. Whiting, and P. P. Roller, "Formaldehyde fixation," *Journal of Histochemistry & Cytochemistry*, vol. 33, no. 8, pp. 845–853, 1985.
- [15] M.-T. Vuong, T. Duval, J. Cohen-Adad, and N. Stikov, "On the Precision of Myelin Imaging: Characterizing Ex vivo Dog Spinal Cord with MRI and Histology," in *Proc. Intl. Soc. Magn. Reson. Med. (ISMRM)*, vol. 25, 2017, Abstract 3760.
- [16] A. L. MacKay and C. Laule, "Magnetic Resonance of Myelin Water: An in vivo Marker for Myelin," *Brain Plasticity*, vol. 2, no. 1, pp. 71–91, 2016.
- [17] H.-L. M. Cheng, N. Stikov, N. R. Ghugre, and G. A. Wright, "Practical medical applications of quantitative MR relaxometry," *Journal of Magnetic Resonance Imaging*, vol. 36, no. 4, pp. 805–824, 2012.

- [18] K.-S. Chan and J. P. Marques, "SEPIA—susceptibility mapping pipeline tool for phase images," *NeuroImage*, vol. 227, art. 117611, 2021.
- [19] W. Li, C. Liu, T. Q. Duong, P. C. M. van Zijl, and X. Li, "Susceptibility tensor imaging (STI) of the brain," *NMR in Biomedicine*, vol. 30, no. 4, e3540, 2017.
- [20] E. J. Canales-Rodríguez et al., "Comparison of non-parametric T2 relaxometry methods for myelin water quantification," *Medical Image Analysis*, vol. 69, art. 101959, 2021.
- [21] C. Laule et al., "Myelin water imaging in multiple sclerosis: quantitative correlations with histopathology," *Multiple Sclerosis*, vol. 12, no. 6, pp. 747–753, 2006.
- [22] B. De Leener et al., "SCT: Spinal Cord Toolbox, an open-source software for processing spinal cord MRI data," *NeuroImage*, vol. 145, pp. 24–43, 2017.
- [23] B. B. Avants, N. J. Tustison, G. Song, P. A. Cook, A. Klein, and J. C. Gee, "A reproducible evaluation of ANTs similarity metric performance in brain image registration," *NeuroImage*, vol. 54, no. 3, pp. 2033–2044, 2011.
- [24] P. McCarthy, "FSLeyes," version 1.5.0 [Software], Wellcome Centre for Integrative Neuroimaging (FMRIB), University of Oxford. Zenodo, 2022. doi: 10.5281/zenodo.7038115.
- [25] P. A. Yushkevich et al., "User-guided 3D active contour segmentation of anatomical structures: Significantly improved efficiency and reliability," *NeuroImage*, vol. 31, no. 3, pp. 1116–1128, 2006.
- [26] R. Salomir, B. Denis de Senneville, and C. T. W. Moonen, "A fast calculation method for magnetic field inhomogeneity due to an arbitrary distribution of bulk susceptibility," *Concepts in Magnetic Resonance Part B*, vol. 19B, no. 1, pp. 26–34, 2003.

- [27] J. P. Marques and R. Bowtell, "Application of a Fourier-based method for rapid calculation of field inhomogeneity due to spatial variation of magnetic susceptibility," *Concepts in Magnetic Resonance Part B*, vol. 25B, no. 1, pp. 65–78, 2005.
- [28] S. Wharton, A. Schäfer, and R. Bowtell, "Susceptibility mapping in the human brain using threshold-based k -space division," *Magnetic Resonance in Medicine*, vol. 63, no. 5, pp. 1292–1304, 2010.
- [29] A. D. Elster, "Magic Angle," *MRIquestions.com*.
<https://www.mriquestions.com/magic-angle.html> (accessed Aug. 19, 2026).
- [30] B. Bilgic et al., "Fast image reconstruction with L2-regularization," *Journal of Magnetic Resonance Imaging*, vol. 40, no. 1, pp. 181–191, 2014.
- [31] T. Liu et al., "Morphology enabled dipole inversion (MEDI) from a single-angle acquisition: comparison with COSMOS in human brain imaging," *Magnetic Resonance in Medicine*, vol. 66, no. 3, pp. 777–783, 2011.
- [32] J. Cohen-Adad et al., "Open-access quantitative MRI data of the spinal cord and reproducibility across participants, sites and manufacturers," *Scientific Data*, vol. 8, art. 219, 2021.
- [33] T. Prasloski et al., "Rapid whole cerebrum myelin water imaging using a 3D GRASE sequence," *NeuroImage*, vol. 63, no. 1, pp. 533–539, 2012.
- [34] T. M. Shepherd, P. E. Thelwall, G. J. Stanisz, and S. J. Blackband, "Aldehyde fixative solutions alter the water relaxation and diffusion properties of nervous tissue," *Magnetic Resonance in Medicine*, vol. 62, no. 1, pp. 26–34, 2009.
- [35] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[36] S. Lévy, M. Benhamou, C. Naaman, P. Rainville, V. Callot, and J. Cohen-Adad, "White matter atlas of the human spinal cord with estimation of partial volume effect," *NeuroImage*, vol. 119, pp. 262–271, 2015.